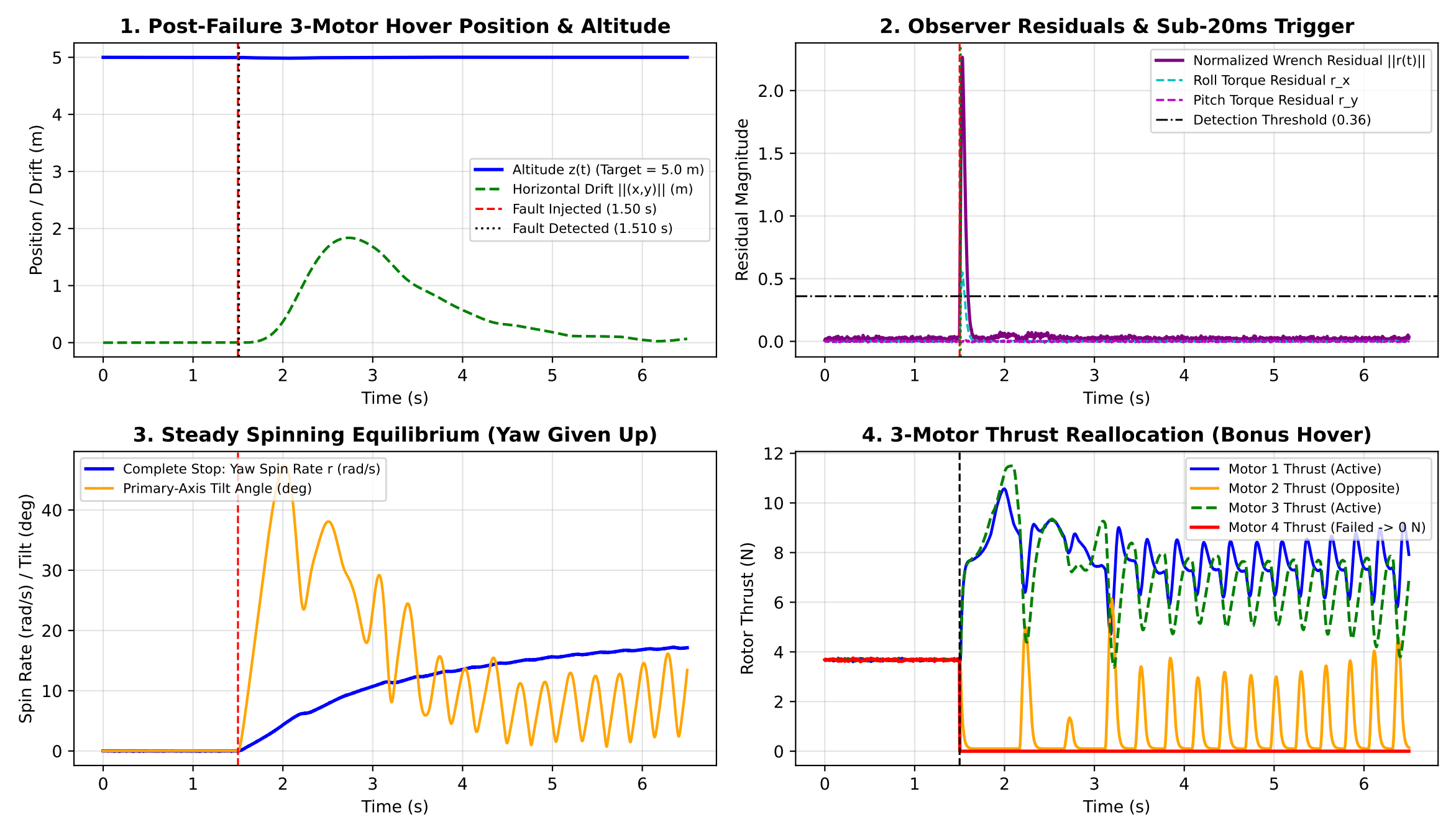

1. Post-Failure 3-Motor Hover Position & Altitude
Position / Drift (m)
Time (s)
Altitude z(t) (Target = 5.0 m)
Horizontal Drift ||(x,y)|| (m)
Fault Injected (1.50 s)
Fault Detected (1.510 s)
2. Observer Residuals & Sub-20ms Trigger
Residual Magnitude
Time (s)
Normalized Wrench Residual ||r(t)||
Roll Torque Residual r_x
Pitch Torque Residual r_y
Detection Threshold (0.36)
3. Steady Spinning Equilibrium (Yaw Given Up)
Spin Rate (rad/s) / Tilt (deg)
Time (s)
Complete Stop: Yaw Spin Rate r (rad/s)
Primary-Axis Tilt Angle (deg)
4. 3-Motor Thrust Reallocation (Bonus Hover)
Rotor Thrust (N)
Time (s)
Motor 1 Thrust (Active)
Motor 2 Thrust (Opposite)
Motor 3 Thrust (Active)
Motor 4 Thrust (Failed -> 0 N)